

Economics 2220

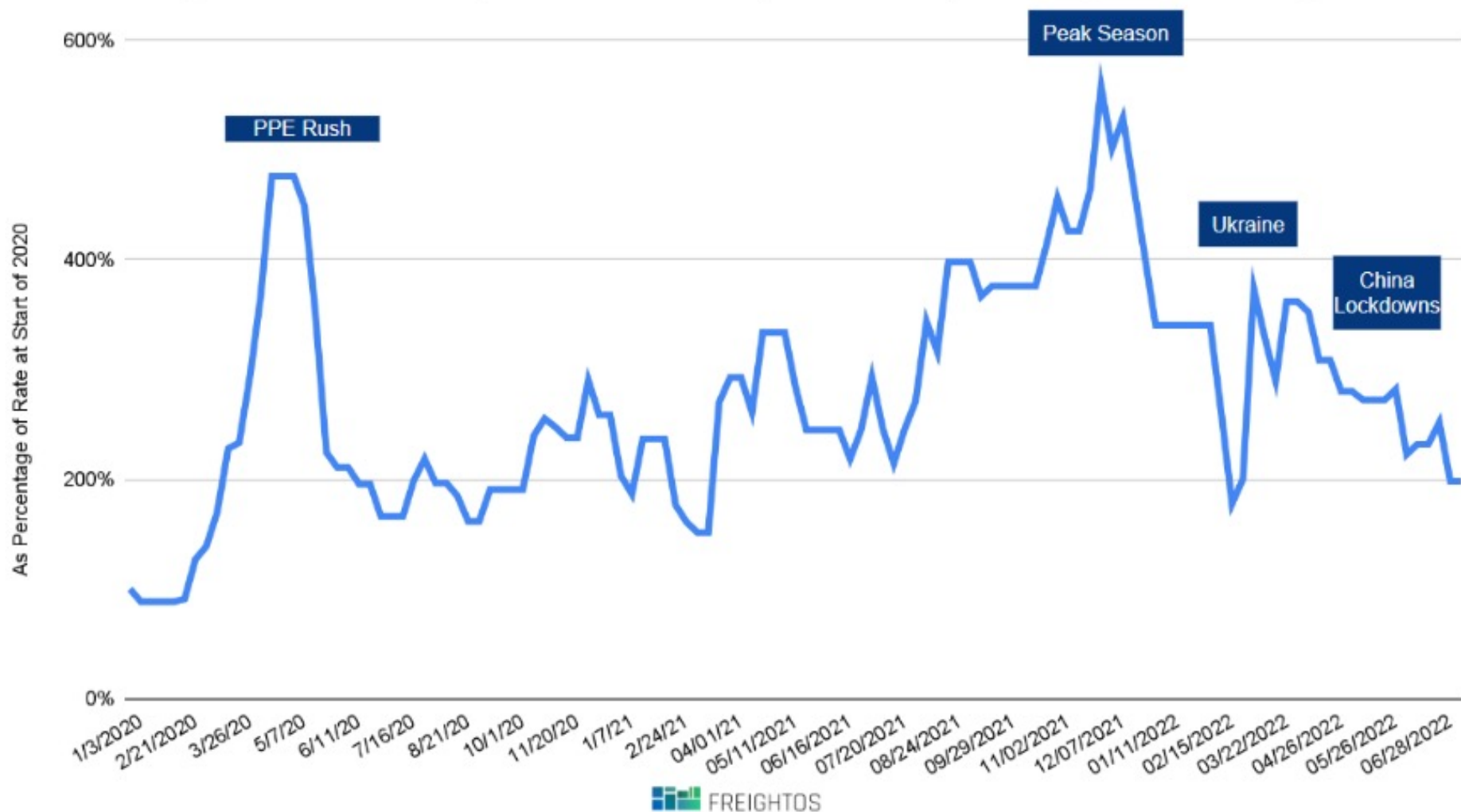
Principles Of Macroeconomics

Professor Zhigang Feng

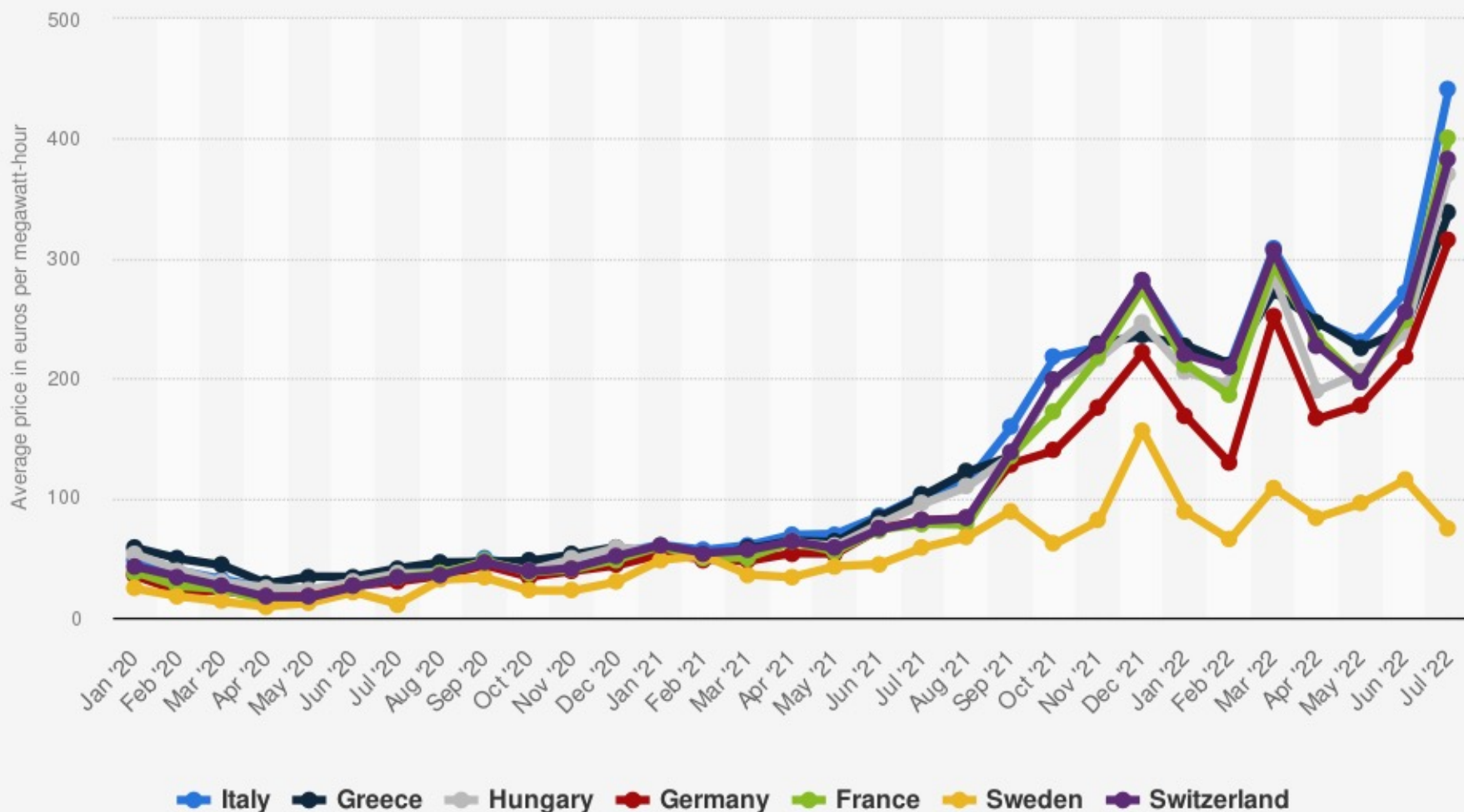
Lecture 4:

Demand and Supply

Freightos.com Marketplace - China-US (West Coast) Air Cargo Rate Changes



Average monthly electricity wholesale prices in selected countries in the European Union (EU) from January 2020 to July 2022 (in euros per megawatt-hour)



► What you will learn in this chapter

- What is a competitive market? How is it described by the supply and demand model?
- What is a demand curve? What is a supply curve?
- What's the difference between movements along a curve and shifts of a curve?
- How do supply and demand curves determine a market's equilibrium price and equilibrium quantity?
- If there's a shortage or surplus, how does price moves the market back to equilibrium?

COMPETITIVE MARKETS

A **competitive market** has many buyers and sellers of the same good or service, none of whom can influence the price.

The **supply and demand model** is a model of how a competitive market behaves.

DEMAND



What made him buy it?

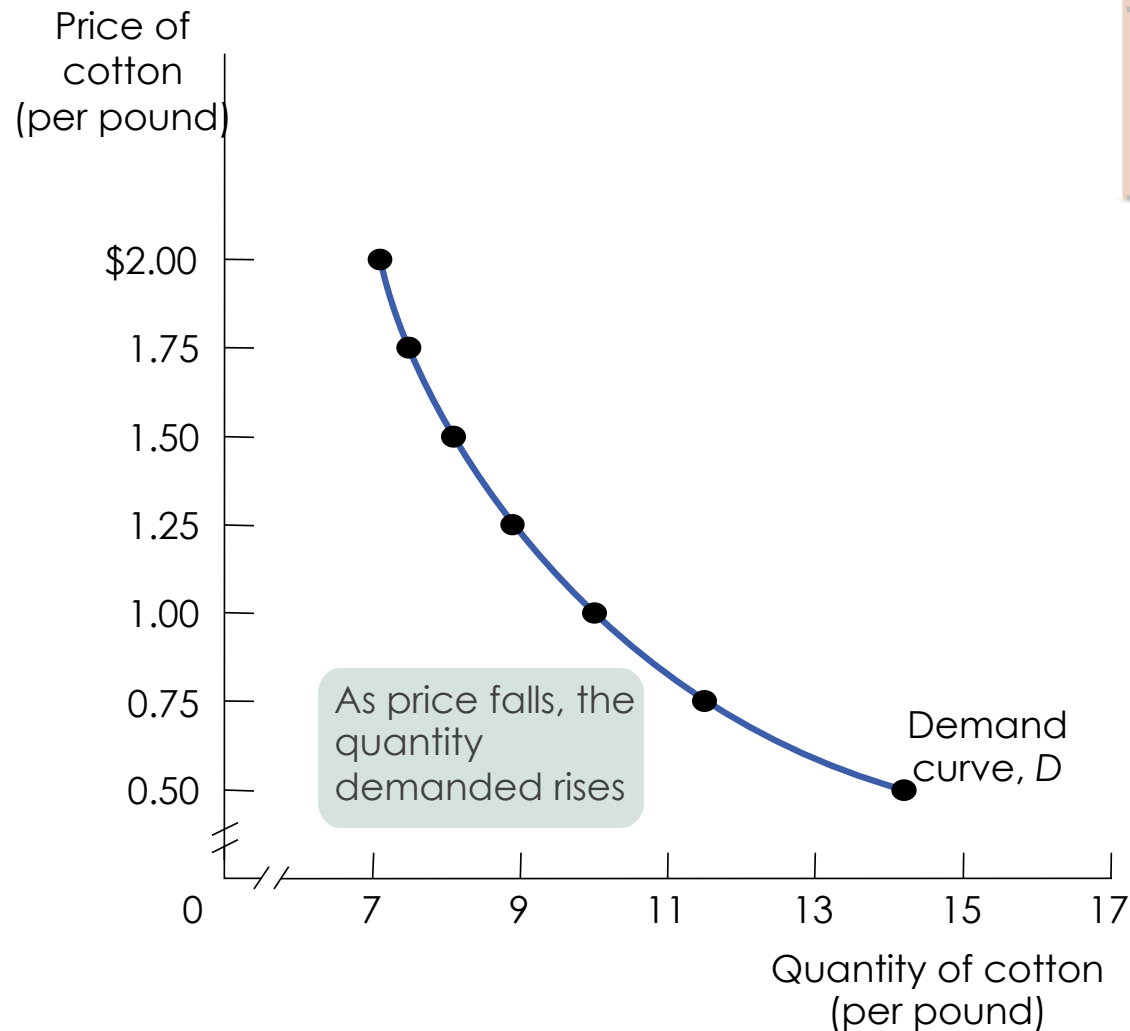
Demand represents the behavior of buyers.

A *demand curve* shows the quantity demanded at various prices.

The *quantity demanded*: the quantity that buyers are willing (and able) to purchase at a particular price.



THE DEMAND SCHEDULE AND THE DEMAND CURVE



Demand Schedule for Cotton	
Price of cotton (per pound)	Quantity of cotton demanded (billions of pounds)
\$2.00	7.1
1.75	7.5
1.50	8.1
1.25	8.9
1.00	10.0
0.75	11.5
0.50	14.2

Don't be sloppy in terminology:

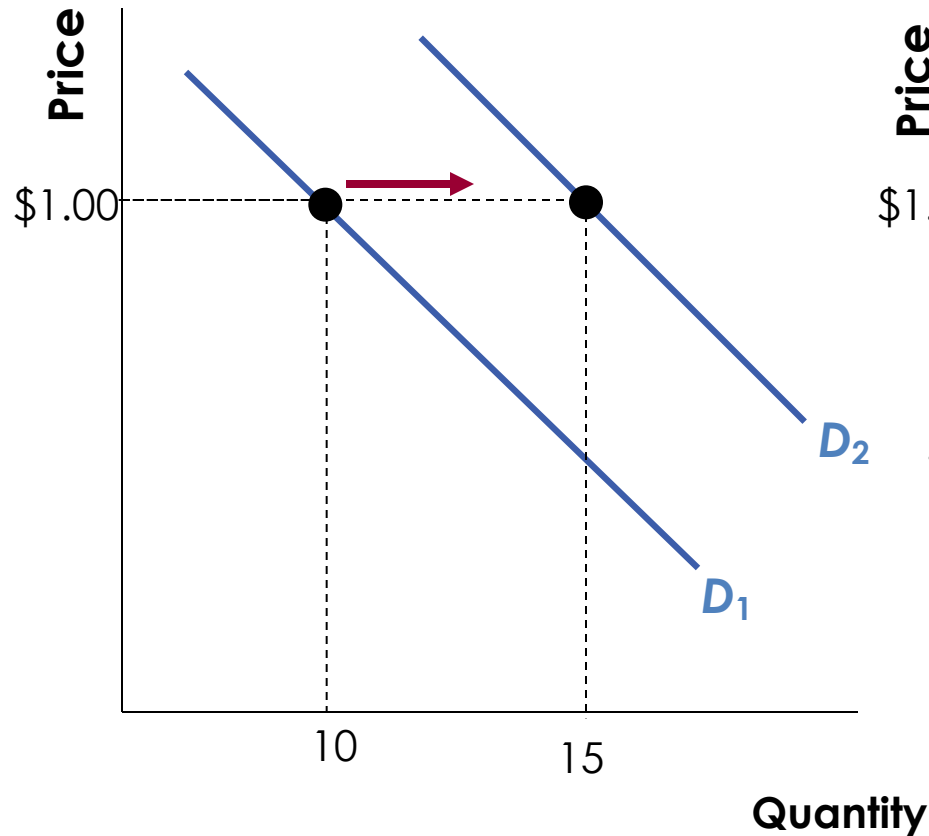
a “*change in demand*”



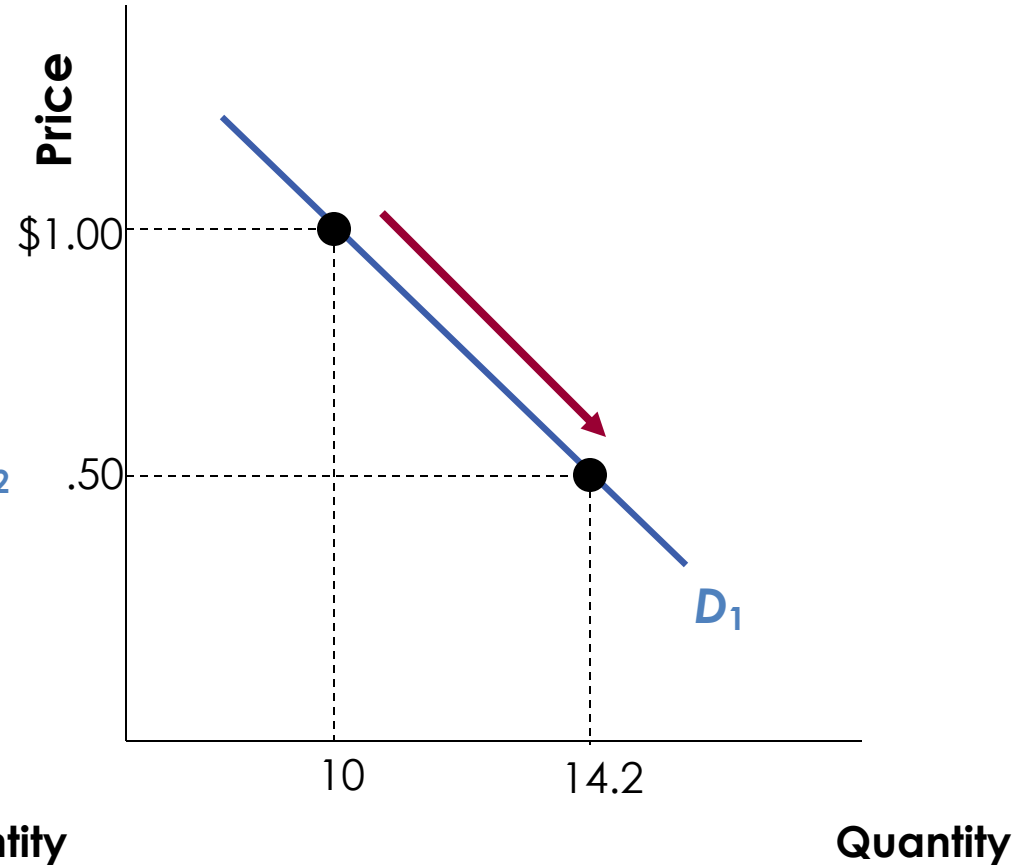
a “*change in quantity demanded*”

CHANGES IN DEMAND VS. CHANGES IN QUANTITY DEMANDED

Change in **demand**



Change in **quantity demanded**



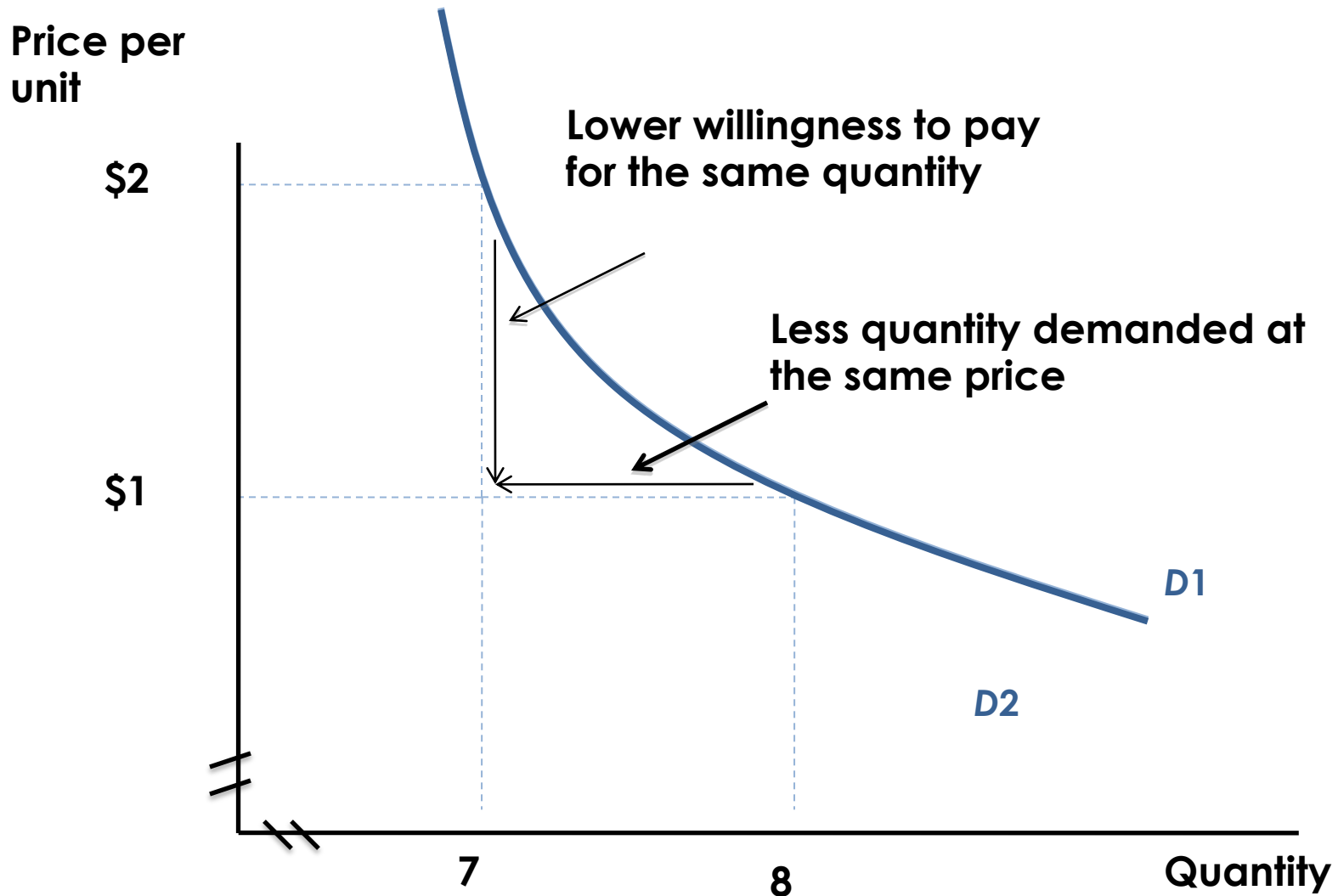
THE DEMAND SCHEDULE AND THE DEMAND CURVE



The Law of demand: a higher price for a good or service leads people to demand a smaller quantity.

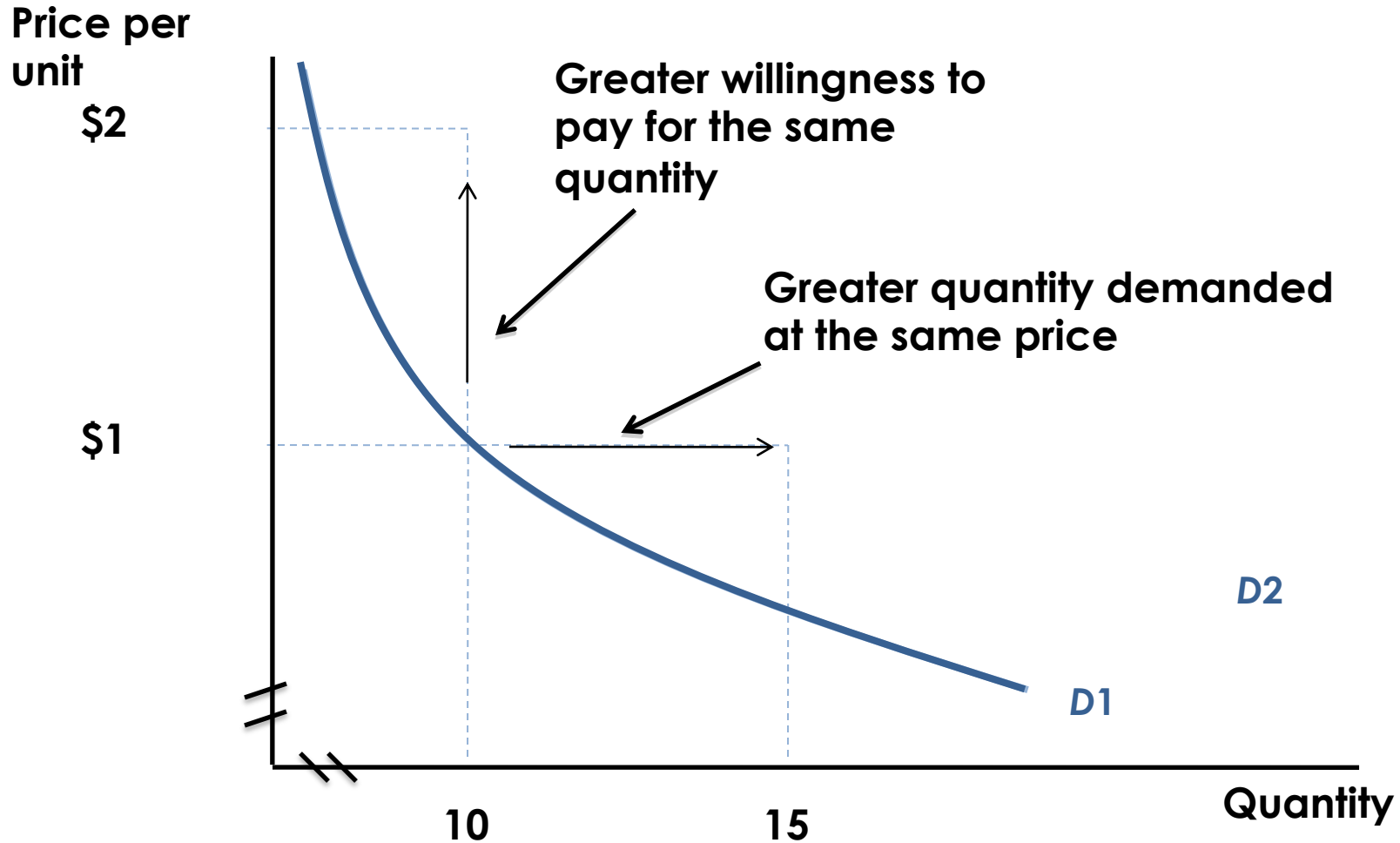
SHIFTS OF THE DEMAND CURVE

A decrease in demand is a leftward shift.



SHIFTS OF THE DEMAND CURVE

An increase in demand is a rightward shift.



UNDERSTANDING SHIFTS OF THE DEMAND CURVE

Important demand shifters:

1. Changes in the **prices of related goods or services**
2. Changes in **income**
3. Changes in **tastes**
4. Changes in **expectations**
5. Changes in the **number of consumers**

CHANGES IN PRICE OF RELATED GOODS: SUBSTITUTES

Two goods are *substitutes* if a decrease in the price of one leads to a decrease in demand for the other (or vice versa).

What happens to the demand for travel in Hawaii if the (perceived) safety cost of traveling to Mexico increases?



CHANGES IN PRICE OF RELATED GOODS: COMPLEMENTS



Two goods are **complements** if a decrease in the price of one good leads to an increase in the demand for the other (or vice versa).

Consumers often have to buy goods together.

An **increase in price of gasoline** will decrease the demand for SUVs.

CHANGES IN INCOME

The effect of changes in income on demand *depends on the nature of the good in question.*

A **normal good**: Demand increases when income increases (and vice versa).

An **inferior good**: Demand decreases when income increases (and vice versa).



CHANGES IN TASTES

Tastes and preferences are subjective and vary among consumers.

Seasonal changes or fads have a predictable effects on demand.

What happens to demand for Face Mask in 2020?



CHANGES IN EXPECTATIONS

If consumers have a choice about the timing of a purchase, they buy according to *expectations*.

Buyers adjust current spending *in anticipation of the direction of future prices* in order to obtain the lowest possible price.

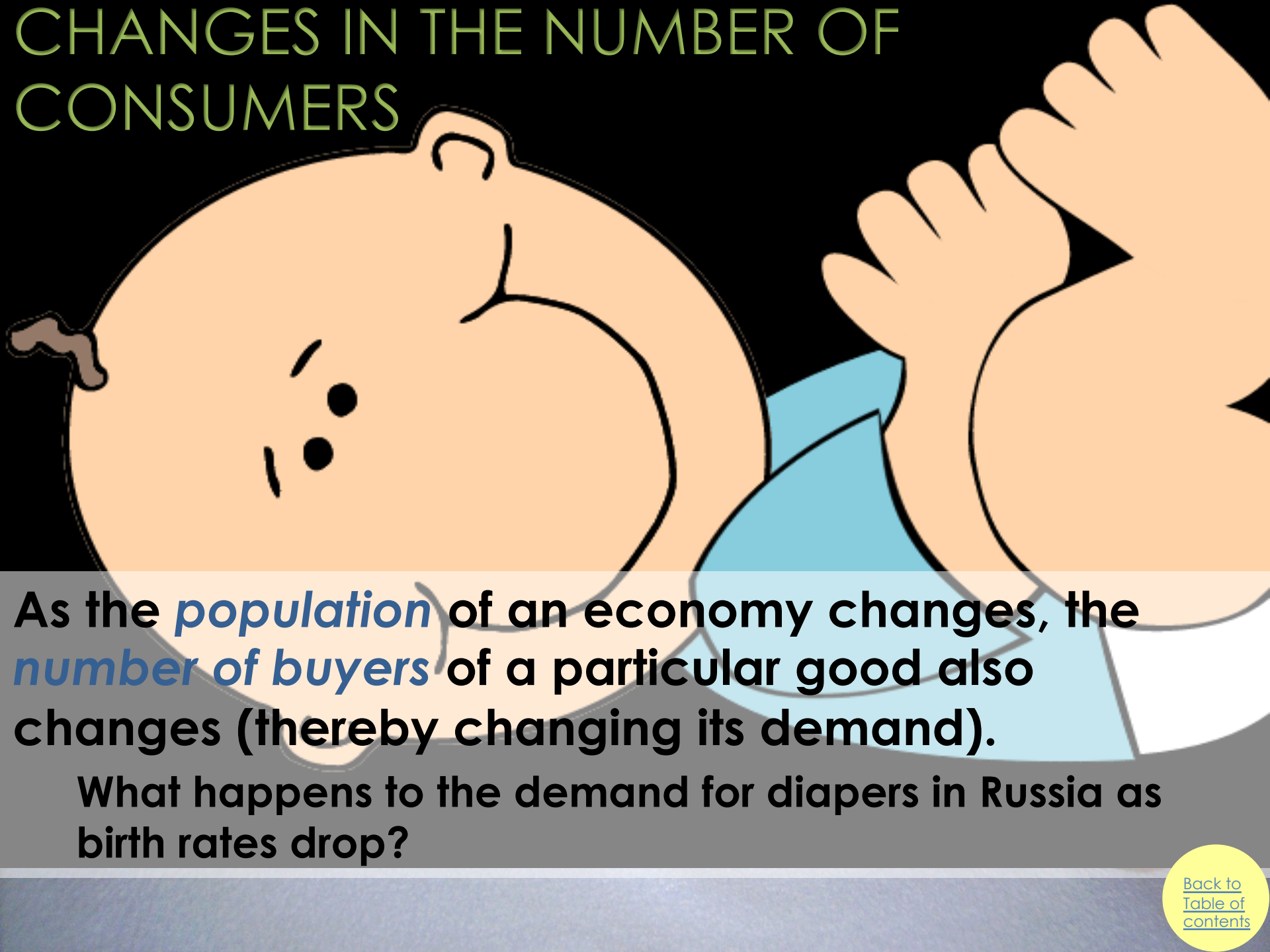
If prices for Xbox one are expected to drop right before Christmas, what will happen to sales during November?

LEARN BY DOING: PRACTICE QUESTION

When the price of petroleum goes up, the demand for natural gas _____, the demand for coal _____, and the demand for solar power _____. (Assume these goods are substitutes for petroleum.)

- a) increases; increases; increases**
- b) increases; increases; decreases**
- c) decreases; decreases; increases**
- d) decreases; decreases; decreases**

CHANGES IN THE NUMBER OF CONSUMERS



As the **population** of an economy changes, the **number of buyers** of a particular good also changes (thereby changing its demand).

What happens to the demand for diapers in Russia as birth rates drop?

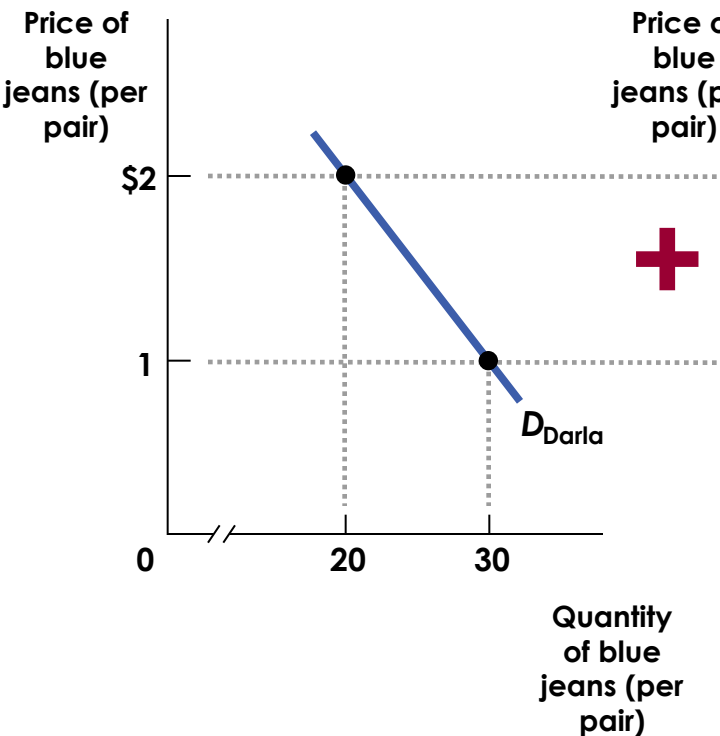
LEARN BY DOING: PRACTICE QUESTION

Which of the following will cause an increase in the demand for autos?

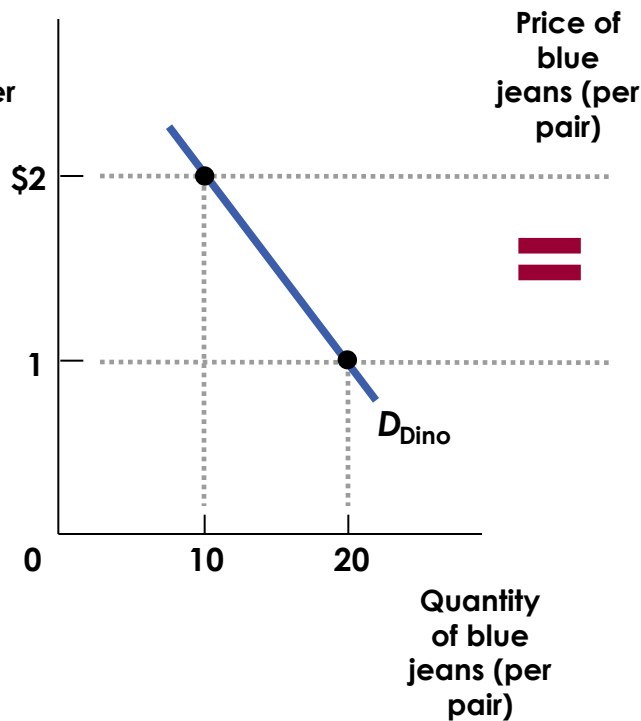
- a) Price of car tires increases because of a Malaysian rubber shortage.
- b) Concrete steel reinforcing rods are replaced by aluminum along the Atlantic coast to prevent rusting.
- c) Gasoline prices drop by 50% when OPEC nations increase production.
- d) McDonald's increases its hamburger production in response to consumer trends.

INDIVIDUAL DEMAND CURVES AND THE MARKET DEMAND CURVE

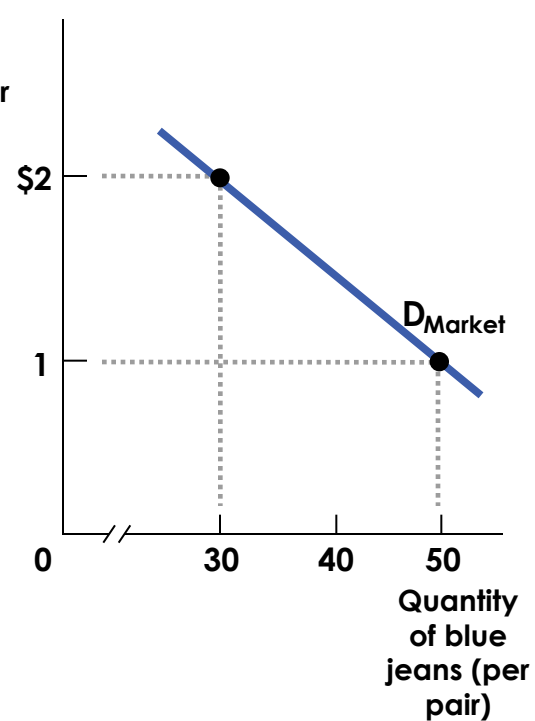
(a) Darla's individual demand curve



(b) Dino's individual demand curve

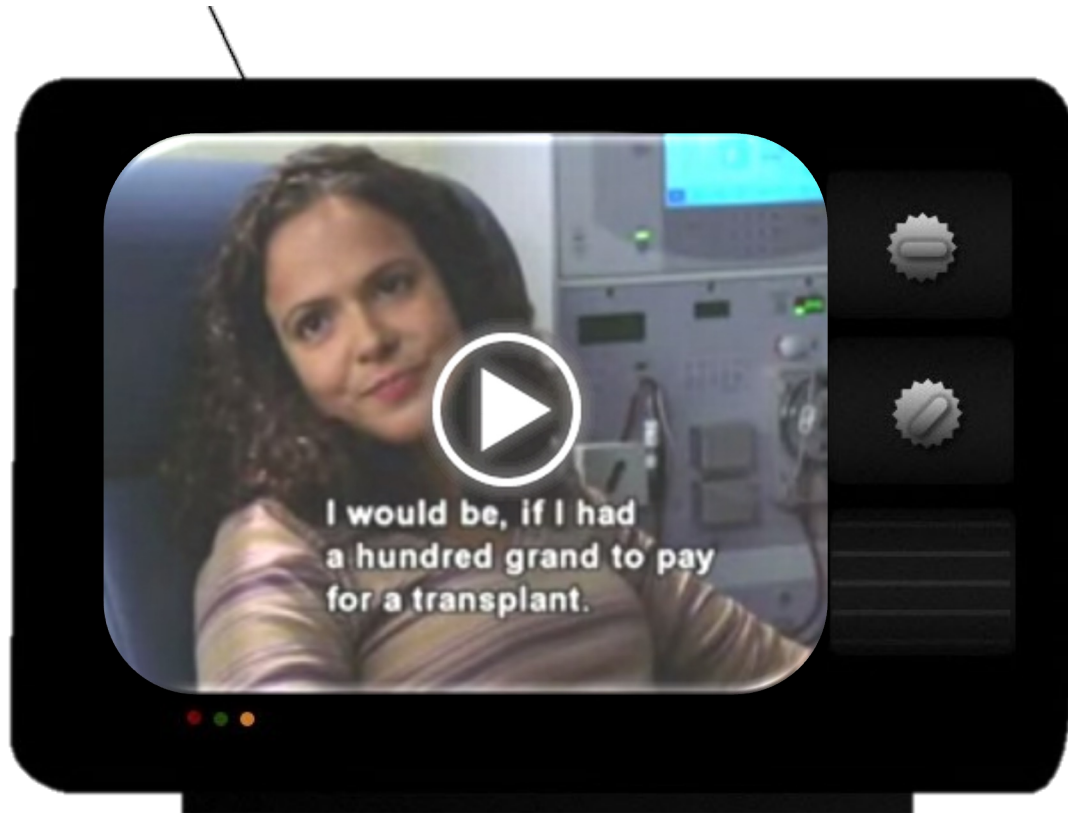


(c) Market demand curve



LEARN BY DOING: APPLICATION VIDEO

Take a look at how the market (legal AND illegal) for kidneys works by clicking [here](#) or on the picture below. (Law and Order SVU) (6:18 minutes)



SUPPLY

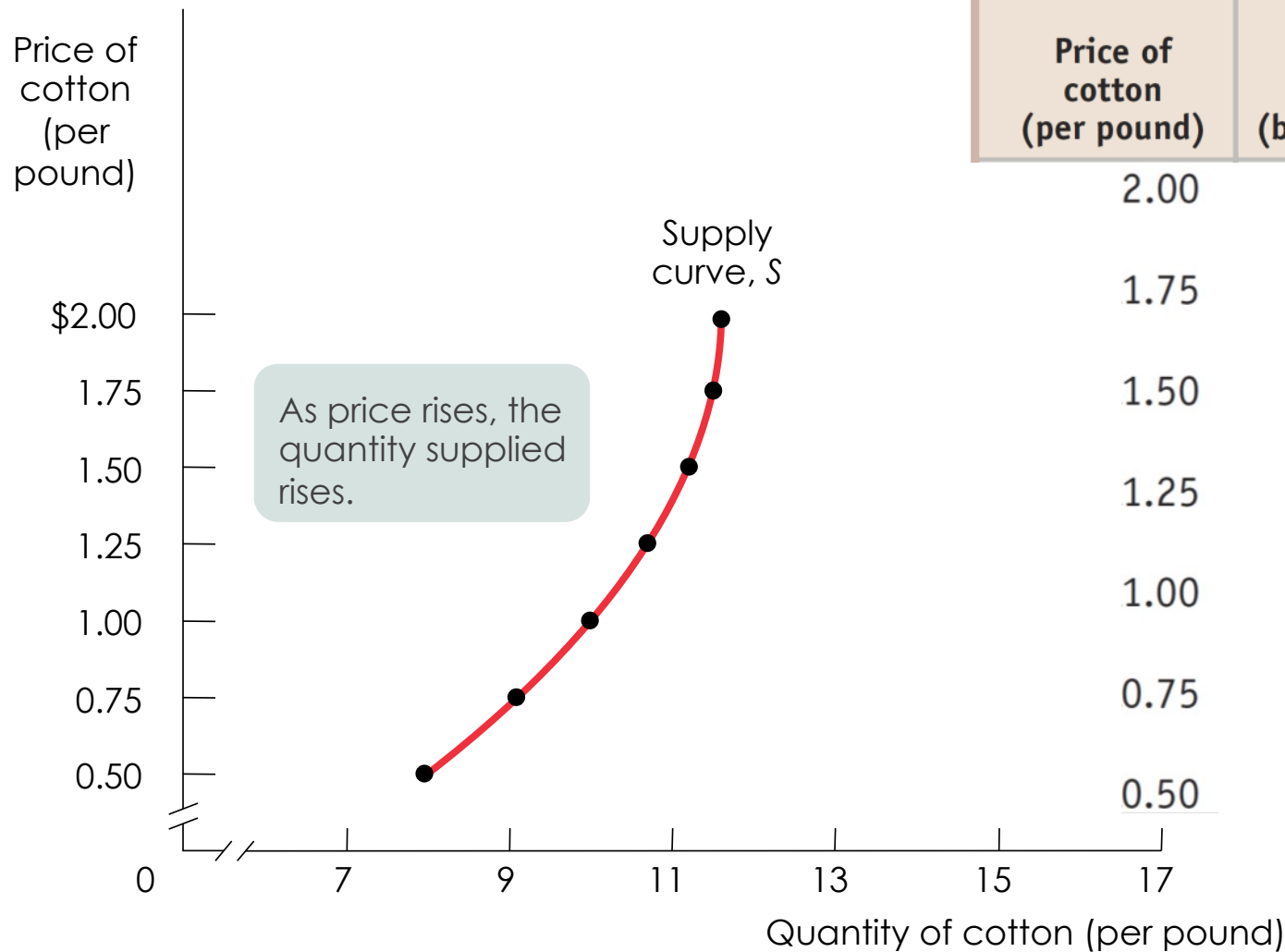
The background of the slide is a photograph of a market stall. It features several blue plastic baskets filled with different types of chili peppers. There are baskets of red chilies, green chilies, and some that are a mix of red and green. The peppers are fresh and vibrant in color. The baskets are arranged in rows, creating a visually appealing display of the product being discussed.

Supply represents the behavior of sellers.

A **supply curve** shows the quantity supplied at various prices.

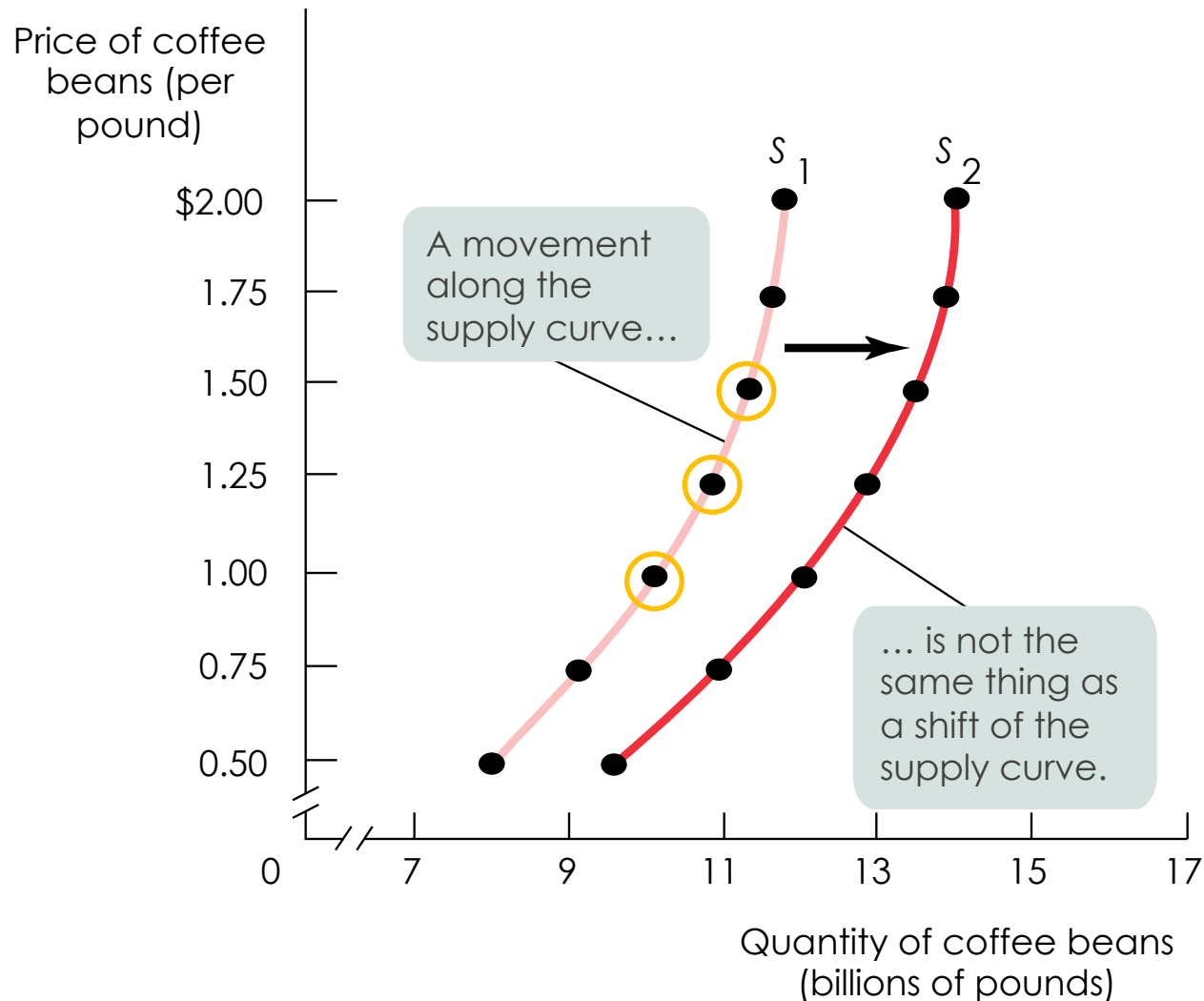
The **quantity supplied** is the quantity that producers are willing and able to sell at a particular price.

THE SUPPLY SCHEDULE AND THE SUPPLY CURVE

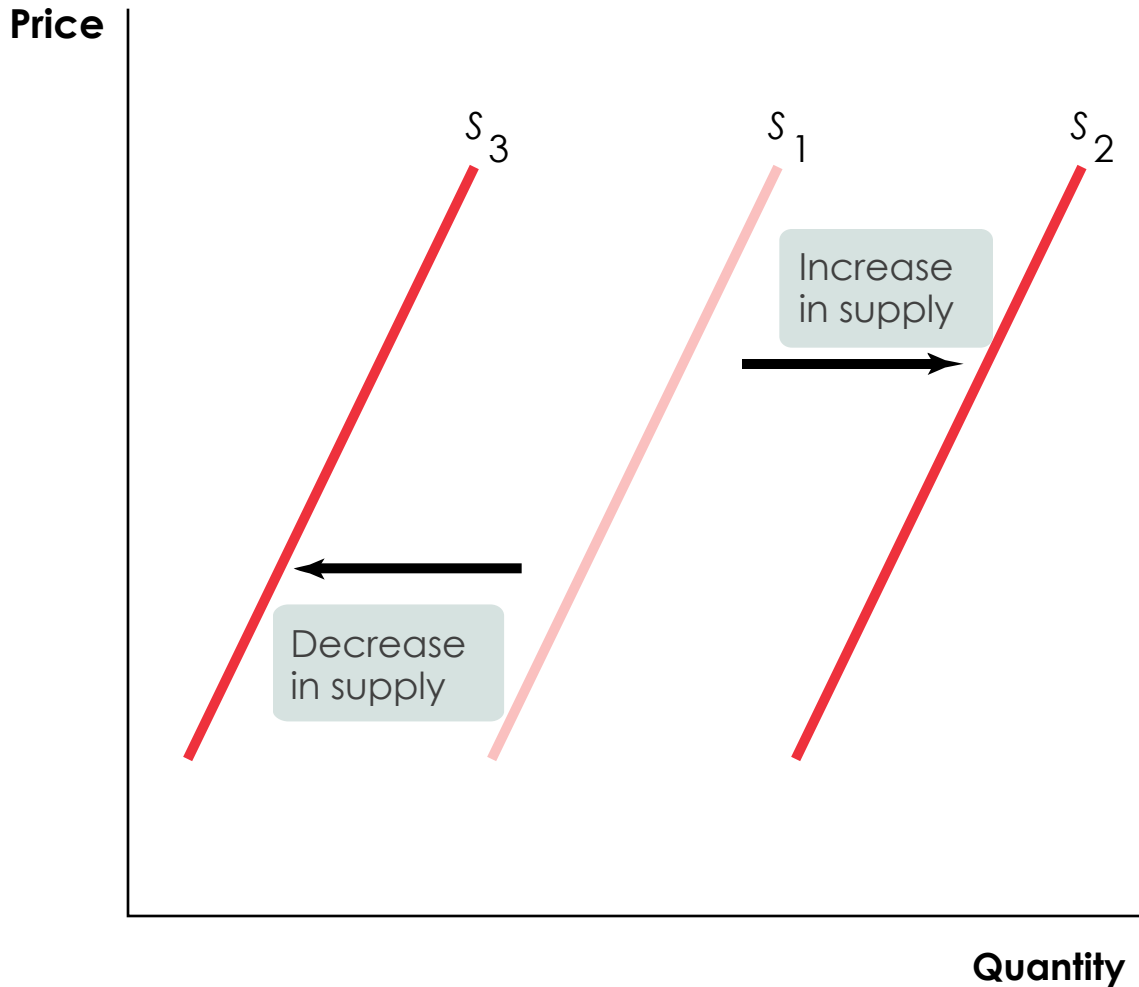


Supply Schedule for Cotton	
Price of cotton (per pound)	Quantity of cotton supplied (billions of pounds)
2.00	11.6
1.75	11.5
1.50	11.2
1.25	10.7
1.00	10.0
0.75	9.1
0.50	8.0

AN INCREASE IN SUPPLY



SHIFTS OF THE SUPPLY CURVE



An **increase in supply** means a *rightward* shift of the supply curve.

A **decrease in supply** means a *leftward* shift of the supply curve.

UNDERSTANDING SHIFTS OF THE SUPPLY CURVE

Important supply shifters include changes in:

- 1. input prices.**
- 2. the prices of related goods or services.**
- 3. technology.**
- 4. expectations.**
- 5. the number of producers.**

CHANGES IN INPUT PRICES

A decrease in the price of an input (all else equal) *increases profits* and encourages more supply (and vice versa).

What will happen to the number of new businesses if the **government reduces the fees and red tape associated with new business licenses?** What happens if the fees rise?



CHANGES IN INPUT PRICES

A close-up photograph of cotton plants. Several large, fluffy white cotton bolls are in focus, attached to brown, dried leaves and stems. The background is a clear, bright blue sky. The image is used as a background for the text.

When the price of cotton drops, the supply of blue jeans increases.

CHANGES IN THE PRICE OF RELATED GOODS OR SERVICES

Inputs used in production have *opportunity costs*. Sellers will choose to use inputs whose profit is the highest.

Sellers will supply **less of a good** if its **profitability falls** (and vice versa).

There are **substitutes** and **complements** in production processes.

Complement in pork processing? Lard.
Substitute in corn production? Cotton.



CHANGES IN TECHNOLOGY

A combine harvester is shown from a low angle, moving through a field of golden wheat. The sky is overcast with grey clouds. The harvester's black metal structure and yellow accents are visible on the right side of the frame.

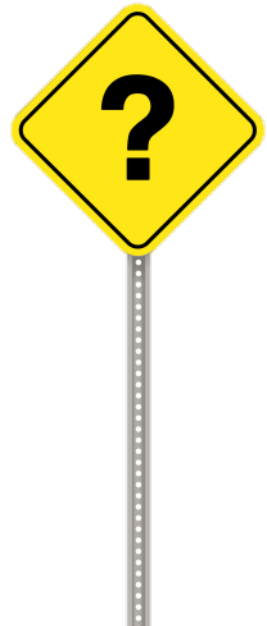
New, better technology makes sellers *willing to offer more at a given price or sell their quantity at a lower price.*

A technological innovation *lowers costs and increases supply.*

CHANGES IN EXPECTATIONS

The **expectation of a higher price for a good in the future** *decreases current supply of the good* – if they can store the good (and vice versa).

Sellers will adjust their current offerings in anticipation of the ***direction of future prices*** in order to obtain the highest possible price.



CHANGES IN EXPECTATIONS



A change in producers' expectations about profitability will affect supply curves.

Windmill production increases as producers expect sales and profitability to increase.

CHANGES IN EXPECTATIONS

The lessons of fidget spinners

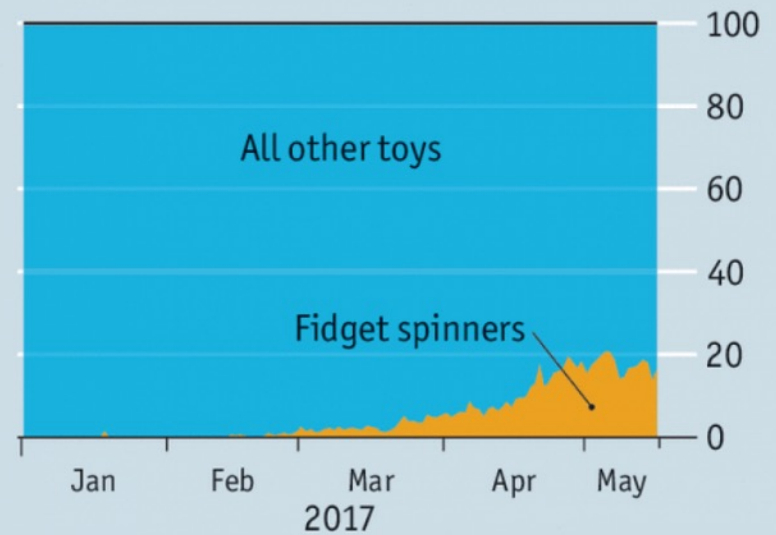
Big retailers usually plan their inventory as much as 18 months ahead of peak seasons such as summer or Christmas.

So it was easy for big retailers to miss the eruption of fidget spinners online. (Subsequently they reacted as well as they could, ordering spinners in by air freight.)



How the world turns

United States, online toy and game unit sales
% of total



CHANGES IN NUMBER OF PRODUCERS

As producers enter and exit the market, the overall supply changes.

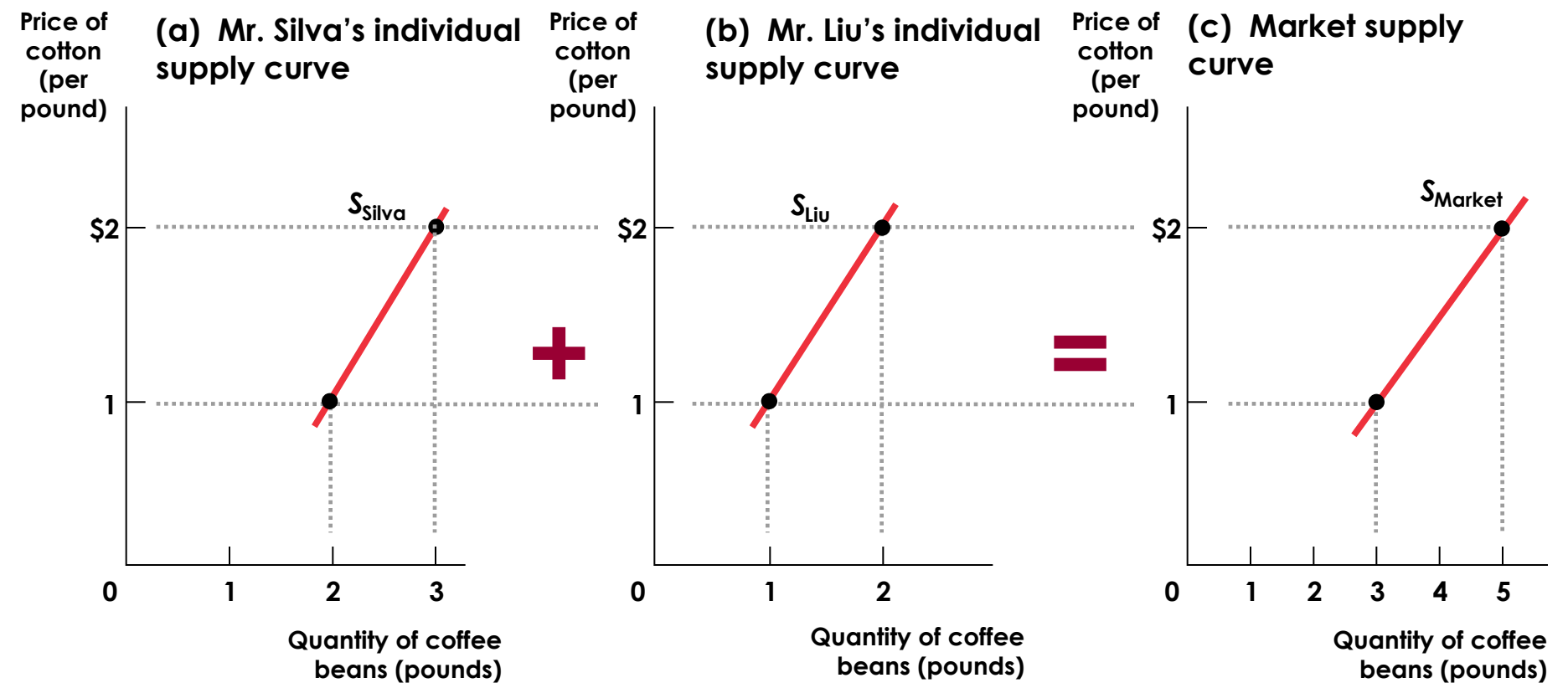
Entry implies more sellers in the market, increasing supply.

Exit implies fewer sellers in the market, decreasing supply.



As more firms enter the solar installation market, the number of solar installations available for sale increases.

THE INDIVIDUAL SUPPLY CURVE AND THE MARKET SUPPLY CURVE



SUPPLY, DEMAND AND MARKET EQUILIBRIUM

When $Q_s = Q_d$ at a certain price, the market is **in equilibrium**.

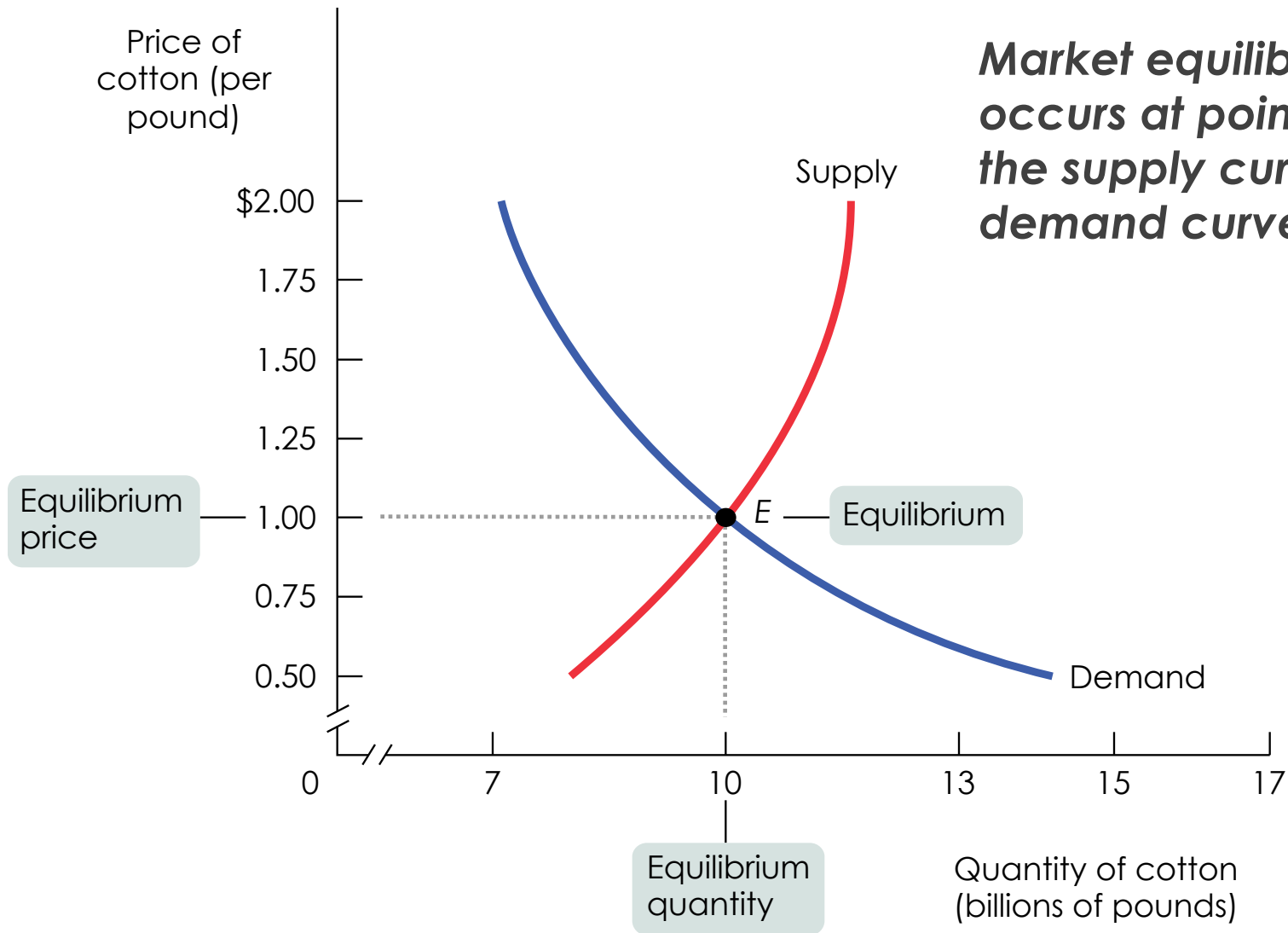
That is, the amount consumers would purchase at this price **is matched exactly** by the amount producers wish to sell.



Because there are middlemen who bring goods to market (and get paid for their efforts), **the price that wholesale cotton farmers (sellers) receive will actually be less than the price paid by retail cotton buyers.**

But have no fear: **the final price paid to retail sellers will match that paid by retail buyers at equilibrium.**

FINDING THE EQUILIBRIUM PRICE AND QUANTITY



WHY DO ALL SALES AND PURCHASES IN A MARKET TAKE PLACE AT THE SAME PRICE?

Where consumers don't have time to compare prices (like a tourist trap), different stores have different prices.



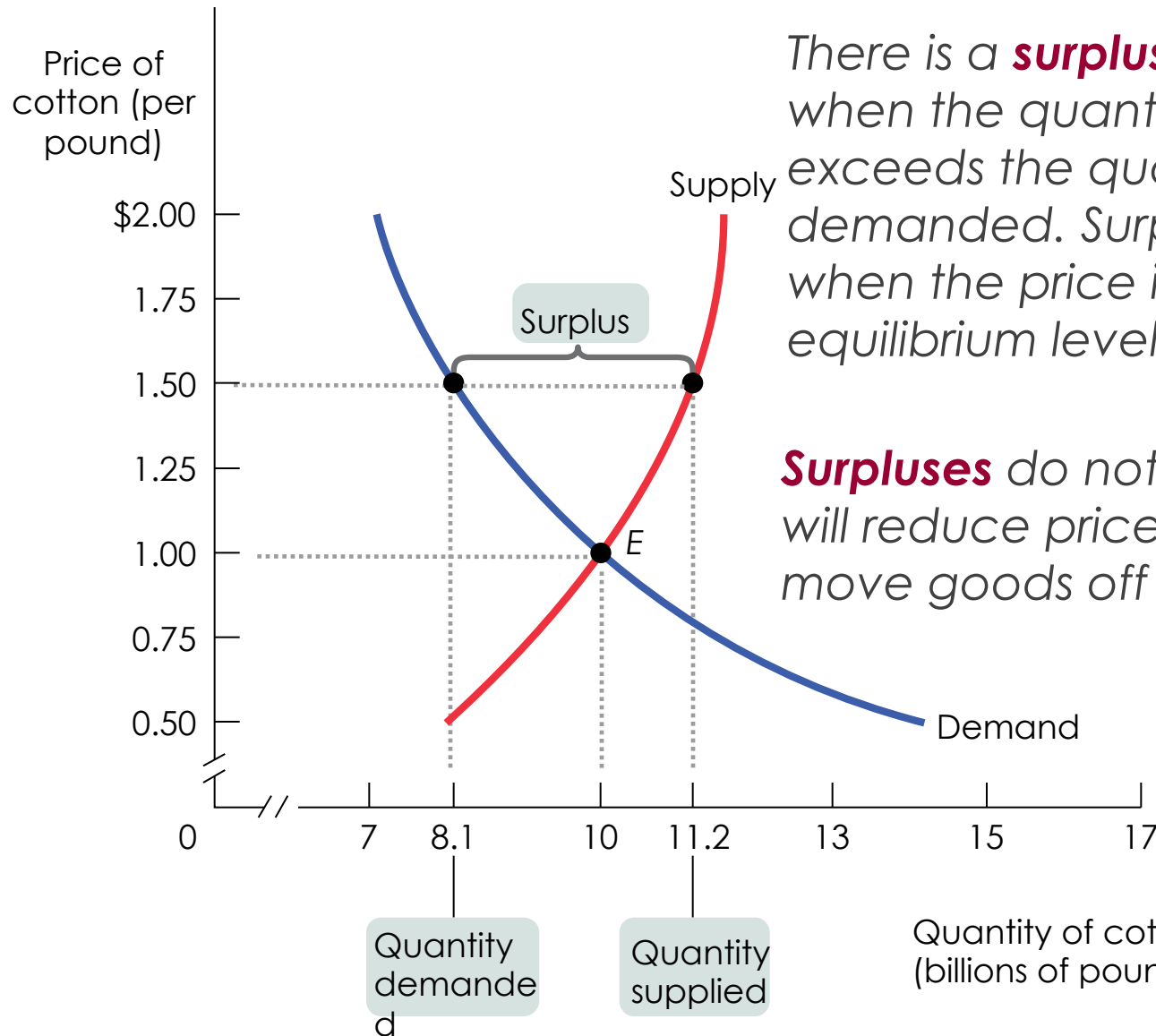
In well-established markets, there is a uniform price.

WHY DO ALL SALES AND PURCHASES IN A MARKET TAKE PLACE AT THE SAME PRICE?

Where consumers don't have time to compare prices (like a tourist trap), different stores have different prices.



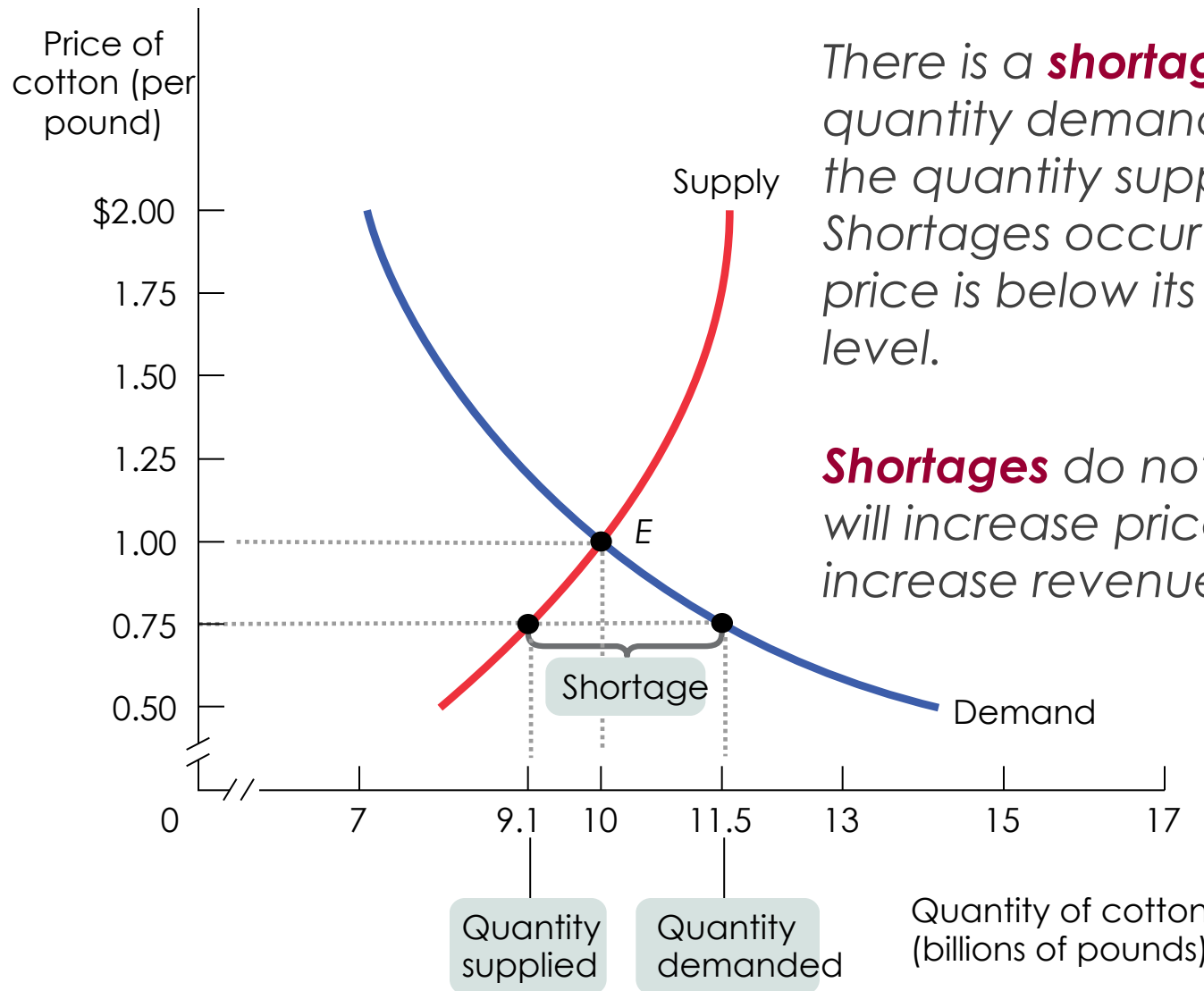
WHY DOES THE MARKET PRICE FALL IF IT IS ABOVE THE EQUILIBRIUM PRICE?



There is a **surplus** of a good when the quantity supplied exceeds the quantity demanded. Surpluses occur when the price is above its equilibrium level.

Surpluses do not last: sellers will reduce price so they can move goods off the shelves.

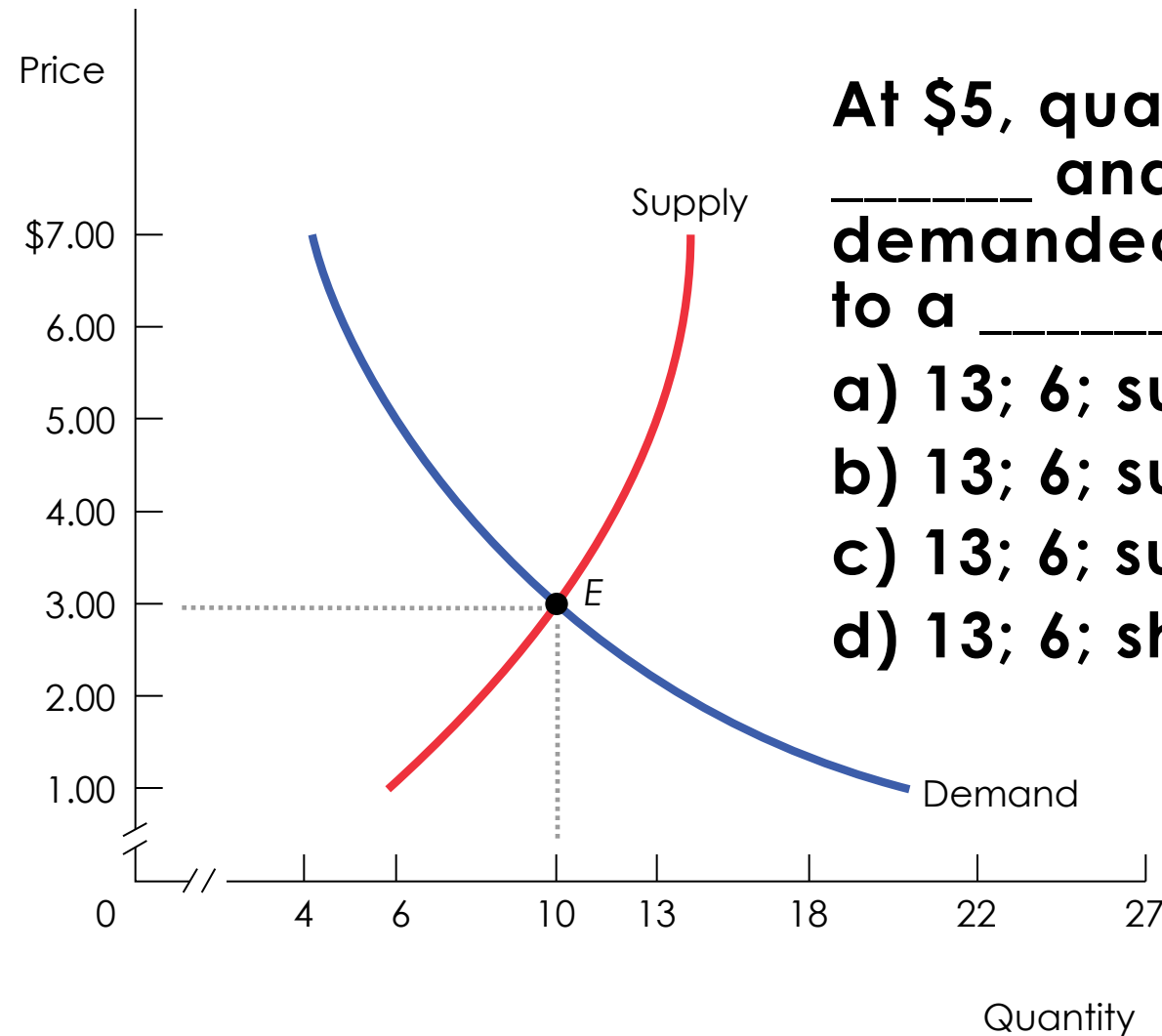
WHY DOES THE MARKET PRICE RISE IF IT IS BELOW THE EQUILIBRIUM PRICE?



There is a **shortage** when the quantity demanded exceeds the quantity supplied. Shortages occur when the price is below its equilibrium level.

Shortages do not last: sellers will increase price to increase revenue.

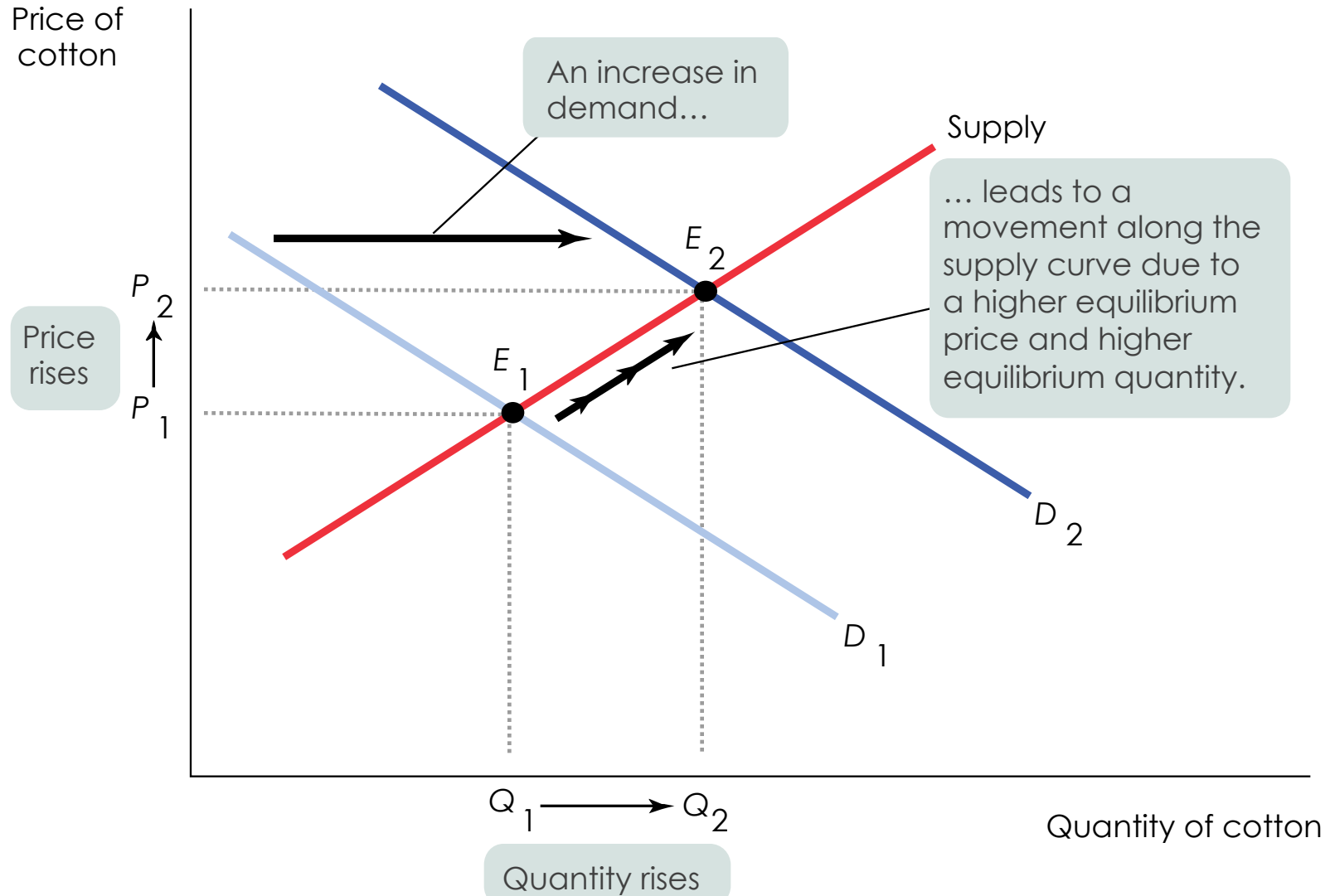
LEARN BY DOING: PRACTICE QUESTION



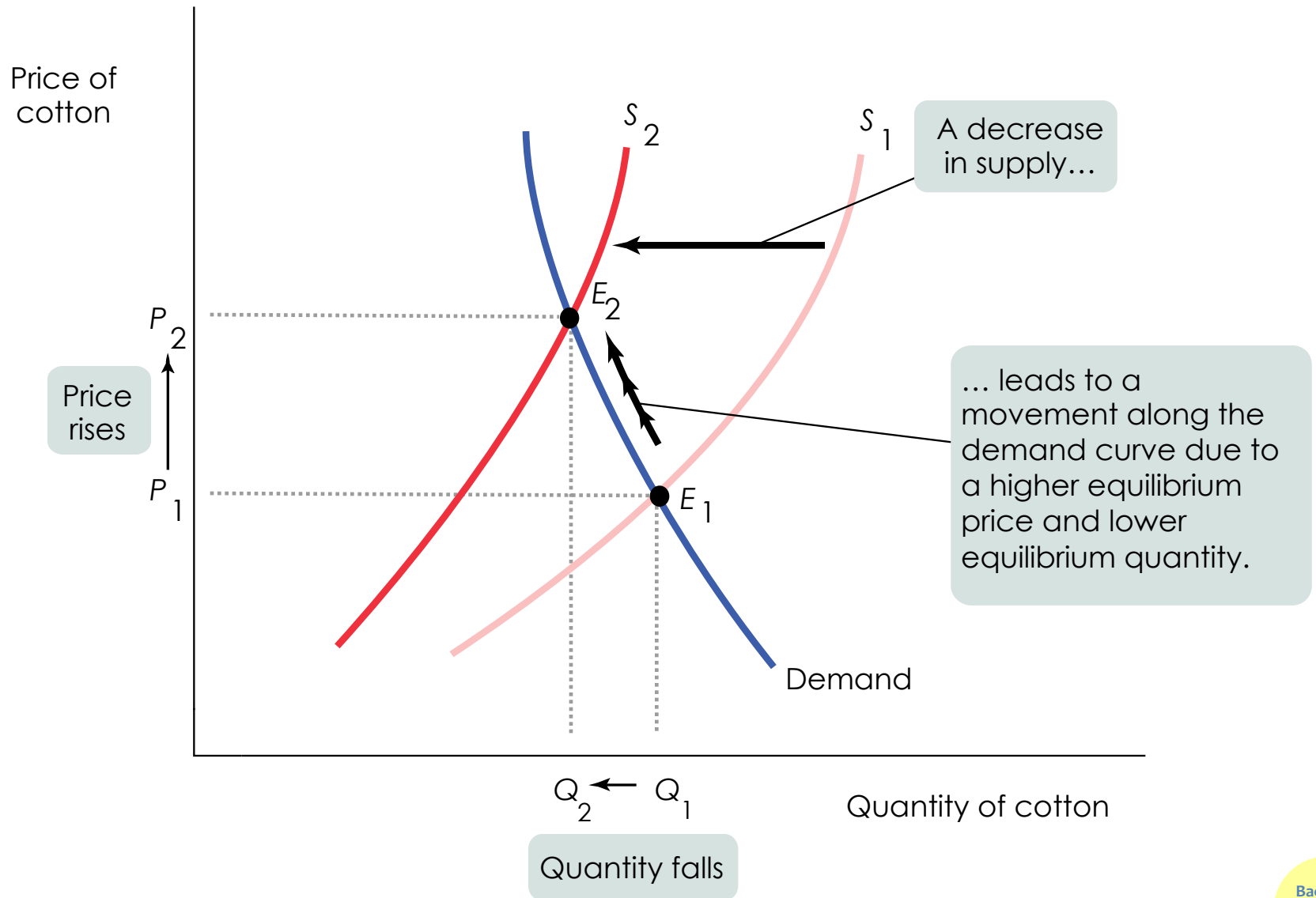
At \$5, quantity supplied is _____ and quantity demanded is _____, leading to a _____.

- a) 13; 6; surplus of 3 units
- b) 13; 6; surplus of 4 units
- c) 13; 6; surplus of 7 units
- d) 13; 6; shortage of 4 units

WHAT HAPPENS WHEN THE DEMAND CURVE SHIFTS



WHAT HAPPENS WHEN THE SUPPLY CURVE SHIFTS



LEARN BY DOING: PRACTICE QUESTION



If garden gnomes regain popularity, what will happen?

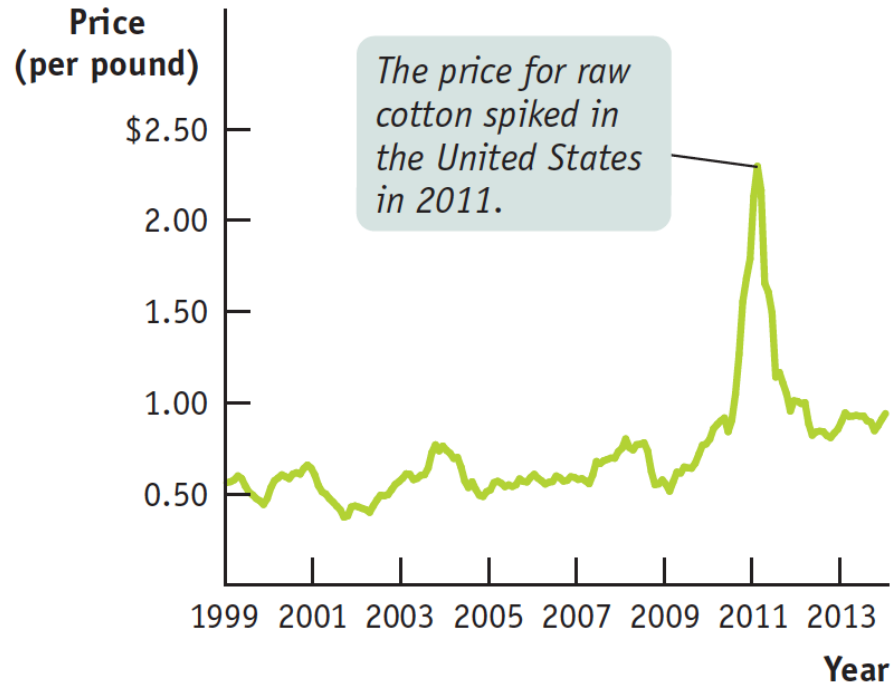
- a) Equilibrium price and quantity both fall.**
- b) Equilibrium price and quantity both rise.**
- c) Equilibrium price falls and quantity rises.**
- d) Equilibrium price rises and quantity falls.**

LEARN BY DOING: PRACTICE QUESTION



If the cost of wood falls, what will happen to equilibrium price and quantity in the violin market?

- a) Both fall.**
- b) Both rise.**
- c) Price falls and quantity rises.**
- d) Price rises and quantity falls.**



Source: USDA.

THE COTTON SPIKE AND CRASH OF 2011

What caused the price spike?

- *Falling supply*
- *Rising demand*
- *Then, in response to higher prices, manufacturers switched to polyester (and eventually supply increased also)*



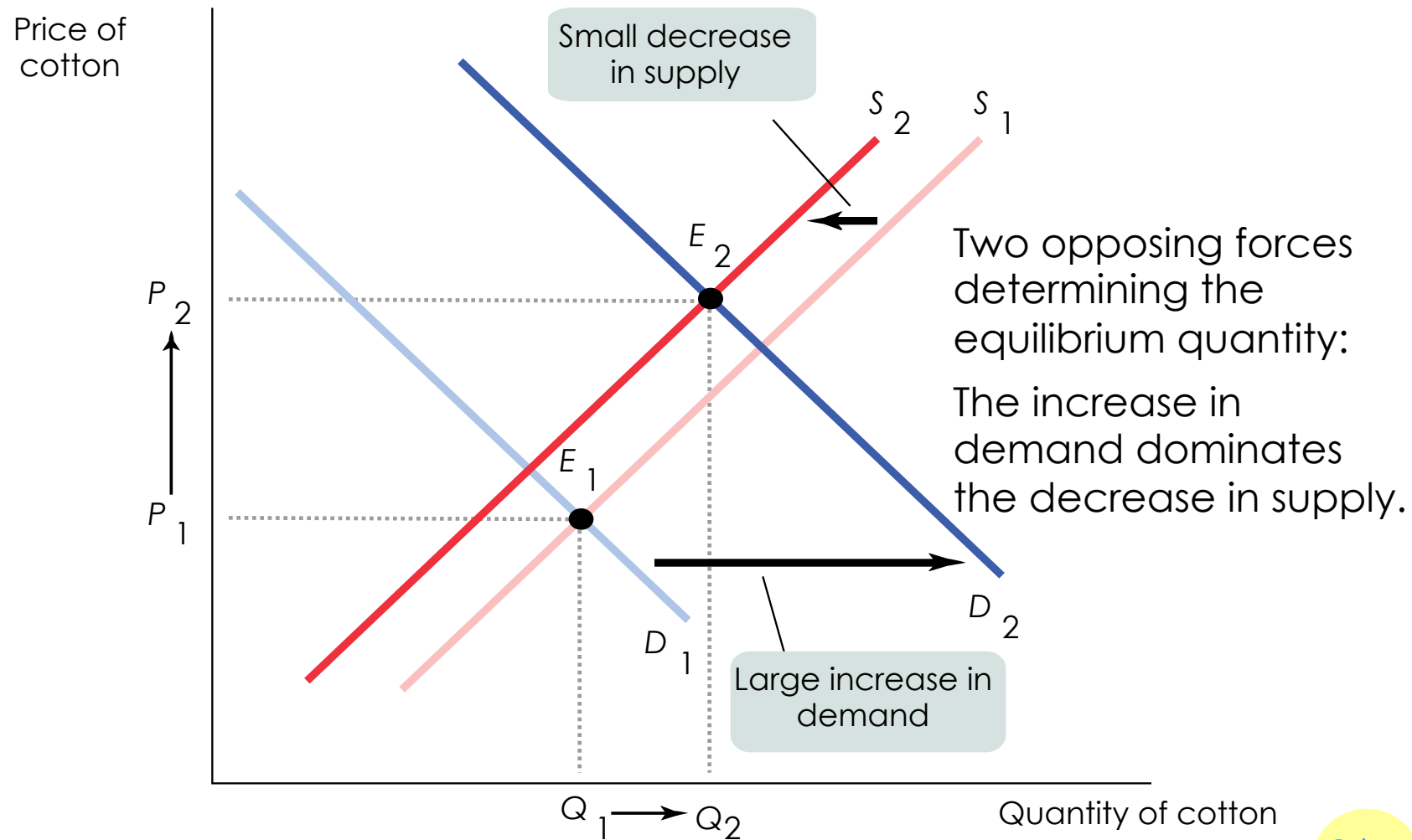
Which curve is it, anyway?

How do you know what caused a price change?

Look at the outcome: if the price and quantity move together, it was a demand change. If they move in opposite directions, a supply change.

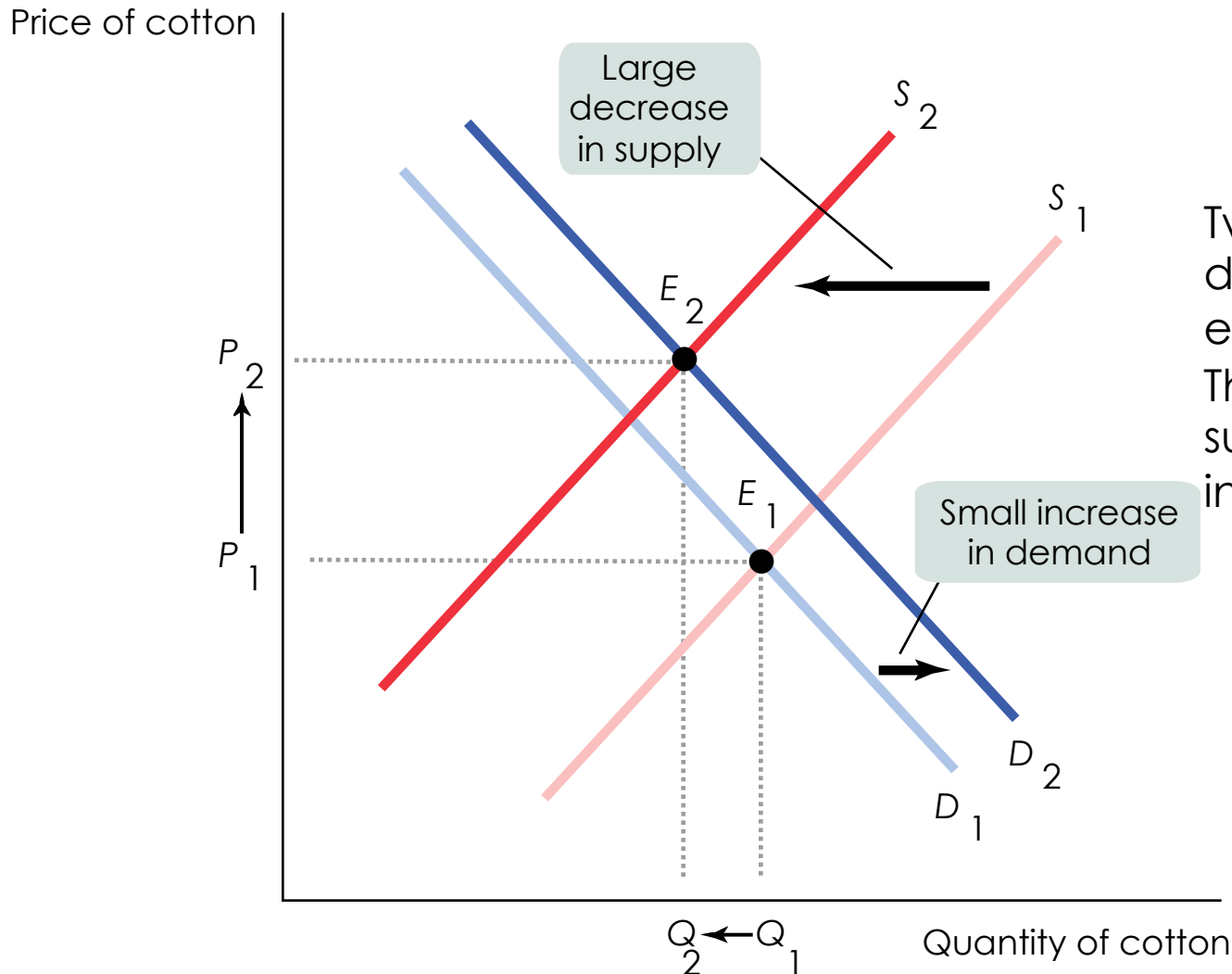
SIMULTANEOUS SHIFTS OF SUPPLY AND DEMAND

(a) **One possible outcome: Price rises, quantity rises.**



SIMULTANEOUS SHIFTS OF SUPPLY AND DEMAND

(b) **Another possible outcome: price rises, quantity falls**



Two opposing forces determining the equilibrium quantity: The decrease in supply dominates the increase in demand.

LEARN BY DOING: PRACTICE QUESTION

Consumers in Mayville consider houses and apartments to be substitutes. There is an increase in the price of houses in Mayville at the same time three new apartment buildings open there. In the market for apartments in Mayville:

- a) the equilibrium price will rise.**
- b) the equilibrium price will fall.**
- c) the equilibrium quantity will rise.**
- d) the equilibrium quantity will fall.**

COMPETITIVE MARKETS—AND OTHERS

When markets are not competitive (as we'll see in future chapters), things get a little more complicated.

For instance, large sellers will have to predict how their production, marketing, and pricing decisions will affect the entire market (and their own profits).

